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DEMOGRAPHIC HISTORY OF KIEV AND PEREYASLAVL (10000BC - 1300AD).

Regional Boundaries, Urban Hierarchy and Population Estimates in the Middle Dnieper Core.

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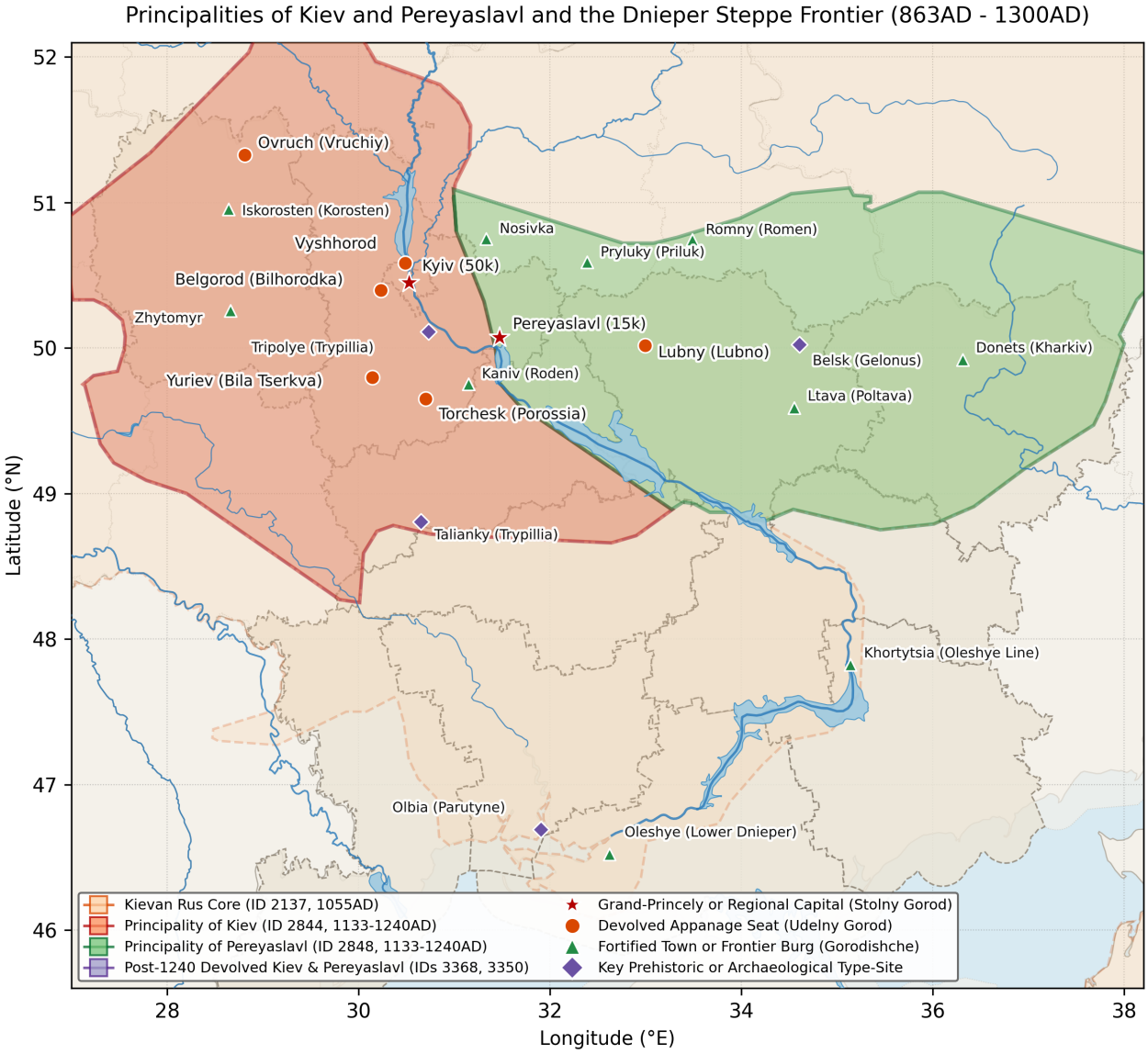


Figure 1. Regional Boundaries and Fortified Settlement Hierarchy of the Principalities of Kiev and Pereyaslavl across the Middle Dnieper Core (10000BC - 1300AD).

Executive Summary and Geographic Scope

This literature review establishes the empirical baseline and settlement-level demographic reconstruction for Region 1 (Kiev and Pereyaslavl) of the Kievan Rus regional sub-series (Stadester and Substrata), book-ended chronologically from the Upper Palaeolithic (10000BC) to the Mongol Conquests (1239-1240AD) and the immediate post-conquest devolved appanage trough (1250AD - 1300AD). Geographically, the region encompasses the Middle Dnieper forest-steppe, the southern Pripyat-Teterev-Uzh Polesian woodlands, the Upper Southern Bug basin and the Lower Dnieper Pontic Steppe corridor (~265 000 km^2 across modern central and southern Ukraine). Within the `saves/atlas_defacto.naissance` historical polity mask, this territory forms the metropolitan core of unified Kievan Rus (2137, 863-1133AD) and subsequently partitions into two sovereign Rurikid principalities and their steppe marches during the era of feudal devolution (1133-1300AD):

1. **Principality of Kiev** (2844 in 1133-1240AD; 3368 and 3465 in 1240-1322AD, ~95 000 km^2): Occupying the Right-Bank Middle Dnieper upland plateau, the Irpin, Stuhna, Ros (*Porossia*), Teterev, Uzh and Sluch river basins (Drevlian Land (Ovruch) Volost), together with the autonomous agricultural buffer of the Bolokhov Land on the Upper Southern Bug and Horyn headwaters.
2. **Principality of Pereyaslavl** (2848 in 1133-1239AD; 3350 and 3466 in 1239-1364AD, ~68 000 km^2): Occupying the Left-Bank Middle Dnieper chernozem forest-steppe across the Trubezh, Supoy, Oster, Sula (*Posullya*), Khorol, Psyol, Vorskla and Upper Seversky Donets basins, functioning as the primary eastern bulwark of Southern Rus against the Pecheneg, Tork and Polovtsian (Cuman) confederations.
3. **Lower Dnieper Emporium and Pontic Steppe Frontier** (~102 000 km^2): Encompassing the Dnieper Rapids (*Porohy*), the Khortytsia-Protolche river crossing, the Kievan maritime enclave of Oleshye (46,62°N, 32,62°E) at the Dnieper-Bug estuary and the pastoral winter pastures (*zimovniki*) of the Black Sea steppe between the Lower Southern Bug and the Molochnaya-Azov littoral.

Across the Holocene and pre-Mongol medieval era, the Middle Dnieper core experienced four distinct demographic cycles prior to the catastrophe of 1239-1240AD: (i) the Eneolithic Trypillia BII-CI proto-urban mega-site phenomenon (c. 3600BC) in the Ros-Southern Bug interfluve, when regional population reached 280,0k [1][2]; (ii) the Scythian forest-steppe hillfort and Greek Pontic colonial expansion (c. 500BC), peaking at 540,0k around the giant rampart complex of Belsk (*Gelonus*, 4 020 ha) and the Milesian polis of Olbia [3][4]; (iii) the Chernyakhov Culture agricultural optimum (c. 300AD), which sustained 720,0k inhabitants across dense open-settlement networks prior to the Hunnic crisis [5]; and (iv) the Kievan Rus urban and rural apogee (1000AD - 1200AD), when the construction of over 950 km of timber-and-earth *Zmiyevy Valy* (Serpent's Walls) and 240 fortified *gorodishcha* supported a regional population of 1,05M in 1000AD, 1,32M in 1100AD and 1,58M in 1200AD [6][7][8][9]. Batu Khan's storming of Pereyaslavl in March 1239AD and the siege of Kiev in November-December 1240AD inflicted an immediate demographic collapse of -53,2% from the 1200AD apogee, reducing the regional population to 740,0k in 1250AD before partial rural and small-town recovery brought the Golden Horde *baskak*-administered appanages to 820,0k by 1300AD [10][11].

Prehistoric and Early Iron Age Settlement (10000BC - 500AD)

Upper Palaeolithic, Neolithic and the Trypillia BII-CI Mega-Site Phenomenon (10000BC - 2000BC)

Human settlement in the Middle Dnieper valley during the terminal Pleistocene (10000BC) was anchored by Epigravettian mammoth-hunter base camps occupying periglacial loess river terraces along the Dnieper, Ros, Supoy and Udai valleys. Excavations at Mezhirich (near Kaniv), Dobranichevka, Gontsy (on the Udai) and Kirillovskaya (in the Podol district of Kiev) reveal circular mammoth-bone dwellings housing winter aggregations of 30 - 60 hunter-gatherers [1][12]. Across the Middle and Lower Dnieper basin, seasonal band territories support a reconstructed Upper Palaeolithic population of 14,0k (14 000) at 10000BC. Following the Mesolithic-Neolithic transition (6000BC), represented by the Bug-Dniester Culture on the Right Bank and the Dnieper-Donets Culture along the Dnieper Rapids (Nikolskoye, Vovnigi, Dereivka) and Left-Bank tributaries, mixed foraging, fishing and incipient stockbreeding communities expanded regional carrying capacity to 42,0k (42 000).

During the Middle and Late Eneolithic (4200BC - 3200BC), the forest-steppe interfluve between the Southern Bug, the Hnylyi Tikych and the Ros-Dnieper line became the demographic centre of gravity of Old Europe with the rise of the Cucuteni-Trypillia Culture (Stages BII and CI, peaking around 3600BC). Geomagnetic surveys and excavations directed by V. A. Kruts, M. Yu. Videiko and international teams have documented a cluster of concentric proto-urban “mega-sites” (*gigant-poseleniia*) in the Cherkasy-Uman plateau that functioned as fortified agro-towns during periods of inter-group competition and steppe encroachment [1][2][13]:

- **Talianky** (48,80°N, 30,53°E, Stage CI, c. 3700-3500BC): Covering 450 ha with ~2 700 geomagnetic house anomalies arranged in concentric ovals; accounting for synchronous house-use horizons (75-80% contemporaneous occupation at 5 - 7 persons per two-storey *ploshchadka* dwelling), Talianky housed 15,0k (15 000) inhabitants at 3600BC.
- **Maidanetske** (48,82°N, 30,69°E): Enclosing 200 ha with ~2 000 - 2 300 structures arranged in four concentric rings and public assembly buildings (*mega-structures*), supporting 10,0k (10 000) inhabitants.
- **Nebelivka** (48,64°N, 30,56°E): A 260 ha Stage BII mega-site with a perimeter ditch, two-storey temple complex and ~1 450 dwellings, supporting 9,0k (9 000) inhabitants.
- Dobrovody (48,77°N, 30,38°E, 250 ha, 8,5k), Sushkivka (120 ha, 4,5k), Vesely Kut (150 ha, 5,0k), Kosenivka (120 ha, 4,0k), Chichirkozovka (100 ha, 3,5k) and the eponymous Trypillia (Tripolye) complex on the Middle Dnieper bluffs (2,5k), alongside Late Trypillia CII Sofievka and Chapayevka cemeteries on the Kievan Dnieper terraces [1].

Aggregating the Uman-Cherkasy mega-site cluster (~65,0k in nucleated settlements >80 ha), the Right-Bank Middle Dnieper and Teterev agricultural villages (~145,0k) and Early Usatovo and Lower Dnieper pastoral communities (~70,0k), the regional population reached an Eneolithic optimum of 280,0k (280 000) at 3600BC. With the aridification of the late fourth millennium BC and the expansion of Yamnaya (Pit-Grave), Catacomb and Middle Dnieper Corded Ware pastoralists by 2000BC, nucleated mega-sites dissolved into mobile steppe kurgan communities and smaller riverside settlements (such as the copper-working and horse-breeding centre at Dereivka and fortified river bluffs at Pustynka), stabilising regional population at 195,0k (195 000) in 2000BC.

Scythian Forest-Steppe Giant Hillforts, Greek Poleis and the Chernyakhov Optimum (500BC - 500AD)

In the Early Iron Age (700BC - 300BC), the synergistic trade of forest-steppe grain, hides, wax and slaves for Milesian Greek wine, olive oil and metalwork through the Lower Bug-Dnieper estuary transformed the Middle Dnieper into the agricultural heartland of Herodotus's "Scythian ploughmen" (*Skolotoi*) and "Geloni and Budini" [3][14]. This era produced two monumental urban phenomena at the 500BC benchmark:

1. **Forest-Steppe Giant Hillforts** (*Gorodishcha-Giganty*): On the Left Bank Vorskla watershed, the Belsk Hillfort (identified by B. A. Shramko with Herodotus's Gelonus, 50,09°N, 34,63°E) enclosed 4 020 ha within 33,8 km of earthen ramparts uniting the Western (72 ha), Eastern (65,2 ha) and Kuzeminsk (15,4 ha) citadels, housing bronze-casting, iron-smelting and Greek pottery workshops with an estimated population of 12,0k (12 000) [3]. On the Right Bank and Left-Bank Sula-Trubezh rivers stood a ring of massive fortified centres: Trakhtemyriv on the Kaniv Dnieper bend (500 ha, 4,5k), Khotiv on the southern outskirts of Kiev (31 ha, 2,5k), Motronyn in the Cherkasy Cold Ravine (200 ha, 4,0k), Pastyrskoye (25 ha, 2,5k), Nemirov on the Southern Bug (110 ha, 4,0k), Basovka on the Upper Sula (170 ha, 3,0k) and Knyazha Hora (1,8k) [14].
2. **Lower Bug-Dnieper Pontic Greek Colonies**: Founded by Milesian colonists in the mid-7th to early 6th centuries BC, Olbia (Pontic Olbia (Parutino)) (46,69°N, 31,90°E, 50 ha stone-walled upper and lower city plus a 100+ rural *chora* network along the Berezan-Bug-Dnieper liman) reached 15,0k (15 000) inhabitants in the urban core at 500BC, complemented by the island emporium of Berezan (Borysthenes) (2,5k), Nikonion (2,0k) and Glubokaya Pristan (1,2k) [4].

Together with dense open agricultural *selishcha* across the Kiev-Cherkasy plateau and royal Scythian pastoralist winter complexes in the Lower Dnieper (precursors to Kamenskoye Gorodishche, 1 200 ha, which peaked in the 4th c. BC at 6,0k), the Middle and Lower Dnieper supported 540,0k (540 000) inhabitants at 500BC.

Following the Sarmatian disruption of the 3rd-2nd centuries BC, the Zarubintsy Culture (200BC - 100AD, named after the cemetery near Trakhtemyriv and Pereyaslavl, with key settlements at Pilipenkova Hora at Kaniv, Chodosovka, Liutezh iron-smelting works north of Kiev and Subotiv) maintained a riverine demographic base of 410,0k (410 000) at 0AD alongside the late Scythian stone fortresses of the Lower Dnieper (Zolota Balka, Gavrilovka, Lyubimovka, Znamenskoye) [15]. During the Late Roman Climatic Optimum (250AD - 375AD), the multi-ethnic Chernyakhov Culture introduced heavy iron ploughshares, rotary querns and wheel-thrown grey-ware pottery across more than 1 200 documented unfortified valley settlements in the Kiev, Pereyaslavl and Cherkasy forest-steppe (Obukhiv, Zhukivtsi, Chernyakhiv, Bilozerka, Kompaniiets and Velikaya Bugaevka). Archaeological synthesis by B. V. Magomedov estimates an average village size of 15 - 25 homesteads (90 - 150 inhabitants) spaced every 2 - 4 km along minor streams, driving the regional population to a pre-medieval peak of 720,0k (720 000) at 300AD [5]. The Hunnic invasion of 375AD and the collapse of Pontic Roman trade networks triggered sharp settlement contraction; by 500AD, the Early Slavic Prague-Korchak Culture (in the Right-Bank Polesian and Teterev woodlands) and the Penkovka Culture (associated with the *Antae* along the Ros, Tyasmin, Sula and Psyol) alongside the early hilltop settlement on Starokyivska Hora (Kiev, 1,5k at 500AD) and the metallurgical craft centre of Pastyrskoye (2,5k in the 6th-7th c. AD) sustained 430,0k (430 000) inhabitants [16][17].

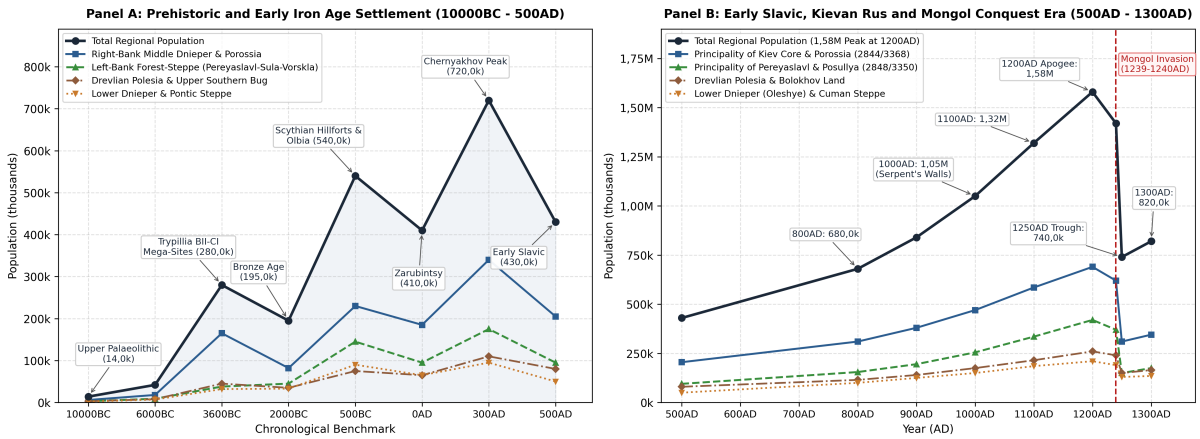


Figure 2. Demographic Trajectory of the Middle Dnieper Core (10000BC - 1300AD)

Benchmark	Right-Bank Kiev & Porossia	Drevlian & Bolokhov Land	Left-Bank Pereyaslavl & Sula	Lower Dnieper & Steppe	Regional Total
10000BC	6,0k	3,0k	3,0k	2,0k	14,0k
6000BC	18,0k	8,0k	9,0k	7,0k	42,0k
3600BC	165,0k	45,0k	38,0k	32,0k	280,0k
2000BC	82,0k	35,0k	45,0k	33,0k	195,0k
500BC	230,0k	75,0k	145,0k	90,0k	540,0k
0AD	185,0k	65,0k	95,0k	65,0k	410,0k
300AD	340,0k	110,0k	175,0k	95,0k	720,0k
500AD	205,0k	80,0k	95,0k	50,0k	430,0k

Table 1. Regional Demographic and Archaeological Data Summary (Table 1).

Early Slavic and Unified Kievan Rus (500AD - 1133AD)

Luka-Raikovetska, Volyntsevo and the Rise of Kiev (800AD - 900AD)

During the 8th and 9th centuries AD (800AD benchmark), Eastern Slavic agrarian colonisation accelerated on both banks of the Middle Dnieper. On the Right Bank, the Luka-Raikovetska Culture established fortified tribal centres (*grady*) and dense rural clusters across the Polanian (*Poliane*), Drevlian (*Drevliane*) and Ulichian (*Ulichi*) lands, including Starokyivska Hora and Zamkova Hora and the earliest dendrochronologically dated log structures in Podol (887AD horizon at Kiev, 5,0k in 800AD, rising to 12,0k in 900AD) [6][18], Iskorosten (Korosten) on the Uzh (1,8k in 800AD), Malyn, Raiky and Monastyrek on theKaniv Dnieper bend (1,2k). On the Left Bank, the Volyntsevo-Romny Culture of the Severians (*Severiane*) occupied the Desna-Seim-Sula-Psyol forest-steppe (Opishnia, Bititsa 2,0k, Romny (*Romen*) 1,2k, Donets 1,0k), operating initially within the tributary orbit of the Khazar Khaganate before Oleg of Novgorod's consolidation of the Dnieper axis in 882AD [19][20]. Regional population recovered to 680,0k

(680 000) at 800AD and reached 840,0k (840 000) at 900AD as Scandinavian-Slavic *druzhina* retinues organised the *polyudie* winter tribute circuit linking the Drevlian, Polanian and Severian river systems to the Byzantine trade treaties of 907AD and 911AD.

Vladimir the Great, Yaroslav the Wise and the Serpent's Walls (1000AD - 1100AD)

The conversion of Rus under Vladimir Svyatoslavich (980-1015AD) and the architectural and legal consolidation under Yaroslav the Wise (1019-1054AD) transformed the demographic landscape of the Middle Dnieper through state-directed frontier engineering and military colonisation. As recorded in the *Primary Chronicle* (*Povest' vremennykh let*, sub anno 988AD and 991AD), Vladimir declared, *"It is not good that there are so few towns around Kiev,"* and systematically constructed concentric defensive lines along the Stuhna, Irpin, Trubezh, Oster, Ros and Sula rivers, staffing them with *"the best men"* relocated from the Slovenes of Novgorod, the Krivichi, the Chud and the Vyatichi [21][22].

Archaeological excavations by M. P. Kuchera across the Serpent's Walls (*Zmiyevy Valy*) demonstrate that between the late 10th and mid-11th centuries, the Kievan state erected over 950 km of continuous timber-laced earthen ramparts (10 - 12 m high, fronted by 5 - 6 m ditches) arranged in six echelons: the Vita-Bobrytsia, Stuhna, Irpin-Teterev, Ros (Porossia), Pereyaslavl-Trubezh and Sula (Posullya) lines [8]. Shielded behind these ramparts, rural *selishcha* density tripled between 950AD and 1100AD:

- **At 1000AD** (1,05M regional total): Kiev expanded from Vladimir's 10 ha *Detinets* (hosting the Church of the Tithes, completed 996AD) and the artisans' quarter of *Podol* to 28,0k (28 000) inhabitants, surrounded by the newly fortified royal residences and garrison towns of Belgorod (110 ha enclosure founded 991AD, 6,0k), Vyshhorod (70 ha, 5,0k), Vasiliev (Vasylkiv) (2,5k), Trepol (Trypillia) (2,0k), Vitichev (1,8k) and the Left-Bank capital of Pereyaslavl-Russky (80 ha fortified complex built 992AD, 8,0k) [6][23].
- **At 1100AD** (1,32M regional total): Under Yaroslav the Wise (who built the 70 ha *City of Yaroslav* enclosed by 3,5 km of 16 m-high ramparts, the Golden Gate and St Sophia Cathedral in 1037AD and founded the Porossia frontier line along the Ros in 1032AD around Yuriev (Bila Tserkva), Torchesk and Korsun) and Vladimir Monomakh (who led the anti-Polovtsian coalitions from Pereyaslavl after the Congresses of Lyubech 1097AD and Dolobsk 1103AD), Kiev reached 40,0k (40 000) inhabitants and Pereyaslavl-Russky reached 12,0k (12 000) [9][24].

Devolved Principalities at Apogee (1133AD - 1240AD)

Following the death of Mstislav the Great (1132AD), the unified realm (2137) devolved into autonomous dynastic lands. In our 1200AD benchmark - the demographic zenith of pre-Mongol Southern Rus - the Middle and Lower Dnieper region attained 1 580 000 (1,58M) inhabitants across 240 archaeologically verified fortified settlements (*gorodishcha*) catalogued by A. V. Kuza [7][23] and O. P. Motsia [9] and over 2 800 open rural villages (*selishcha*).

Principality of Kiev (2844): Metropolitan Core, Right-Bank Ring, Porossia, Drevlian Land and Bolokhov (~950,0k at 1200AD)

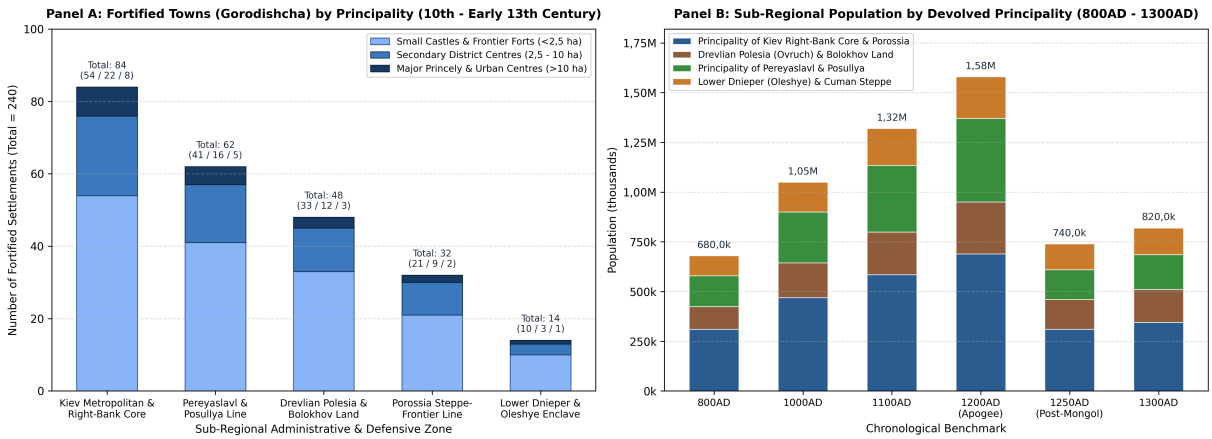


Figure 3. Settlement Hierarchy and Devolved Principality Demography in the Middle Dnieper Core (800AD - 1300AD)

The Kievan Metropolis (45,0k - 50,0k at 1200AD)

Detailed planimetric and stratigraphic reconstructions by P. P. Tolochko [6][12][24], M. A. Sahaydak [18] and H. Yu. Ivakin [10] establish that late 12th- and early 13th-century Kiev comprised a polycentric urban agglomeration spanning 360 - 400 ha (380 ha standard benchmark):

1. **The Upper City** (*Gora*, ~98 ha): Consisting of *Vladimir's City* (10-12 ha, princely *dvor* chambers, stone churches and boyar compounds), *Yaroslav's City* (70 ha, metropolitan cathedral of St Sophia, St Irene, St George and dense residential quarters), *Izyaslav-Svyatopolk's City* (8 ha, St Michael's Golden-Domed Monastery) and *Kopyrev Konets* (8 ha on the northwestern spur).
2. **The Lower City** (*Podol*, 180 - 200 ha): Dendrochronological excavations across *Kontraktova Ploshcha*, *Zhytniy Rynok*, *Verkhniy Val* and *Nyzhniy Val* reveal standardized 300 - 600 m^2 fenced urban homesteads (*usadby*) containing 2 - 3 log dwellings, craft workshops (glass-making, bone-carving, enamelling, shoemaking, smithing) and merchant warehouses (*pirohy*). Allowing 3 500 - 4 000 active homesteads across *Podol* at 6 persons per household yields 22,0k - 25,0k residents in the Lower City alone [18].
3. **Suburban Monastic and Princely Satellites** (~80 - 100 ha): Including the *Kievo-Pecherskaya Lavra* (Caves Monastery), the princely residence of *Berestovo*, *Klov*, *Dorogozhichi*, *Vydubychi*, *Kyrylivska*, *Kitaevo* (35 ha fortified complex on the southern Dnieper hills) and *Gnilets*.

Combining Upper City boyar and clerical households (~15,0k), *Podol* artisan and merchant homesteads (~24,0k), *Kopyrev Konets* and the suburban districts (~9,0k - 11,0k), Tolochko's authoritative estimate of 50,0k (50 000) inhabitants at 1200AD (equivalent to ~131,6 persons/ha across 380 ha) places Kiev among the largest metropolises of 12th-century Europe alongside Constantinople, Paris and London [24][25].

The Right-Bank Kievan Ring and Stuhna-Dnieper Corridor

Protecting the southern and western approaches to Kiev stood a dense network of princely seats, custom ports and fortress-towns (84 fortified *gorodishcha* in the Kiev Core zone: 8 major >10 ha, 22 secondary 2,5-10 ha and 54 small <2,5 ha castles):

- **Belgorod** (Bilhorodka) on the Irpin (50,39°N, 30,23°E, 110 ha *detinets* and *okolny gorod*, episcopal see

and favourite secondary capital of Rurik Rostislavich): 9,0k (9 000) at 1200AD.

- **Vyshhorod** (50,58°N, 30,49°E, 70 ha cliffside citadel and potter's *posad*, shrine of SS Boris and Gleb): 7,0k (7 000).
- **Kaniv** (49,75°N, 31,47°E, 24 ha, strategic Dnieper crossing and assembly point against the Cumans, stone Church of St George 1144AD): 4,0k (4 000).
- Vasiliev (Vasylkiv) on the Stuhna (20 ha, 3,5k), Trepol (Trypillia) (18 ha Dnieper cliff fortress controlling the Stuhna mouth, 3,0k), Roden on *Knyazha Hora* (16 ha craft town near Kaniv, 2,5k), Vitichev (14 ha beacon fortress on the Dnieper, 2,2k), Zarub (1,5k), Chuchin (12 ha, 1,8k), Khalep (1,4k), Svyatoslavl, Ivan-Gorod (1,5k), Tumasch (1,4k), Kitaigorod (1,5k) and Zvenyhorod-Kievan (1,6k) [7][9].

The Porossia Steppe-Frontier Line (*Chorni Klobuky* Federates)

Along the Ros River valley (80 km south of Kiev), the Kievan grand princes settled semi-nomadic Turkic federates - the Chorni Klobuky ("Black Hoods"), uniting the Torks, Berendei, Kovui, Turpei and remnant Pechenegs - as hereditary border cavalry endowed with fortified towns (32 *gorodishcha* in the Porossia zone: 2 major, 9 secondary, 21 small) [9][22]:

- **Torchesk** (49,68°N, 30,63°E, identified with the 45 ha *Olshanitsa (Sharki)* complex on the Gorokhovatka-Ros): Capital of the Black Hoods and appanage seat of Mstislav the Bold (1193-1207AD), supporting 6,0k (6 000) inhabitants at 1200AD.
- **Yuriev** (Bila Tserkva) (49,79°N, 30,11°E, 28 ha *Zamkova Hora* and *posad*, episcopal seat of the Bishop of Yuriev): 4,5k (4 500).
- Korsun on the Ros (49,42°N, 31,26°E, 16 ha, 2,8k), Boguslavl (49,55°N, 30,87°E, 15 ha, 2,5k), Volodarev (Volodarka) (1,8k), Rostovets (1,8k), Dveren (1,5k), Polksten (1,5k), Bakhmutov (1,2k), Sakkov (1,1k) and Borovyi (1,0k).

Together, the Right-Bank Kiev Core and Porossia frontier supported 690,0k (690 000) inhabitants at 1200AD.

Drevlian Polesia (Ovruch Volost) and the Bolokhov Land (~260,0k at 1200AD)

Northwest and southwest of the Kievan core lay two distinct sub-regions encompassing 48 fortified *gorodishcha* (3 major >10 ha, 12 secondary 2,5-10 ha, 33 small <2,5 ha):

1. **Drevlian Land** (Ovruch)-Korosten Polesia: Centred on Ovruch (Vruchiy) (51,32°N, 28,80°E, 25 ha, brick Church of St Basil built by Rurik Rostislavich c. 1190AD, 4,5k at 1200AD) and Iskorosten (Korosten) (50,95°N, 28,64°E, four-citadel complex on the Uzh, 22 ha, 4,0k). As demonstrated by A. P. Tomashevsky [26], the Ovruch ridge was the sole European source of pink pyrophyllite slate, surrounded by over 40 specialized workshop villages (*Nagorany*, *Kamenka*, *Korenevka*) manufacturing slate spindle-whorls (*priaslitsa*) and sarcophagi exported across all of Rus, Poland and Volga Bulgaria. Secondary Drevlian centres included Vozvyagl (Zviahel) on the Sluch (16 ha, 2,8k), Malyn (1,8k), Kotelnich (2,0k), Zhytomyr (1,8k), Horodsk (*Gorodesk*) (2,0k), Michesk (Radomyshl) (1,6k), Ushchesk (Ushomyr) (1,5k) and Chornobyl (1,5k at the Pripyat-Uzh confluence).
2. **The Bolokhov Land** (Bolokhovskaia Zemlia, 1150-1257AD): Occupying the fertile forest-steppe plateau of the Upper Southern Bug, Sluch and Horyn between the principalities of Kiev, Galicia-Volhynia and the Cuman steppe. Governed by autonomous local elites (*knyaztsi bolokhovskie*), its fortified towns - Bolokhov (Lyubar) (15 ha, 2,5k), Gubin (12 ha, 2,0k), Kudin (1,8k), Kobud (1,6k), Derevich (1,6k), Bozhsky (*Bozhesk*) (2,0k), Mezhibozhye (Medzhybizh) (2,2k) and Dyadkov (1,4k) - were renowned for intensive wheat and millet cultivation [9][27].

Principality of Pereyaslavl (2848): Left-Bank Forest-Steppe and the Posullya Rampart Line (~420,0k at 1200AD)

East of the Dnieper, the Principality of Pereyaslavl (2848) served throughout the 11th to early 13th centuries as the coveted ancestral patrimony of the Monomakhovichi and the primary military shield of Southern Rus [19][28]. Archaeological surveys by Yu. Yu. Morgunov and O. V. Sukhobokov record 62 fortified *gorodishcha* across the Pereyaslavl land (5 major >10 ha, 16 secondary 2,5-10 ha, 41 small <2,5 ha):

- **Pereyaslavl-Russky** (Pereiaslav) (50,07°N, 31,46°E): Located at the confluence of the Alta and Trubezh rivers, the capital comprised a 10 ha *Detinets* (containing Bishop Ephraim's monumental stone Cathedral of St Michael 1089AD, stone episcopal gates, bathhouse and princely palace) and a 70 ha fortified *Okolny Gorod* plus unfortified riverside *posady* (80 ha walled core, ~100 ha total complex), supporting 15,0k (15 000) inhabitants at 1200AD [6][7].
- **The Posullya Defensive Line** (Sula River Basin): Stretching 180 km from the Dnieper estuary of the Sula to the Upper Sula border with the Kursk-Putivl Severians, the Posullya line comprised 18 inter-visible timber-and-earth *gorodishcha* anchored at the mouth by the fortified harbour-citadel of Voin (Voinska Hreblia) (18 ha with an artificial dredged interior ship-harbour, 2,5k at 1200AD), followed upstream by Zheldi (1,4k), Rimiv (14 ha, heroically besieged by Konchak in 1185AD, 2,2k), Lukoml (13 ha, 2,0k), Lubno (Lubny) (16 ha, founded by Monomakh, 3,0k), Khorol (2,0k), Sneporod (12 ha, 1,8k), Ksniatin (13 ha, 2,0k), Sinets (1,5k), Romen (Romny) (15 ha, 2,5k) and Glinsk (1,5k) [28].
- **Interior Trubezh-Supoy-Oster and Eastern Psyol-Vorskla-Donets Basins**: Behind the Sula line lay Priluk (Pryluky) on the Udai (15 ha, 2,8k), Nosivka (2,2k), Oster (Gorodets-Ostersky) (14 ha, founded by Monomakh 1098AD with St Michael's *Yurievska Bozhnytsia*, 2,5k), Baruch (1,8k), Bronnitsa (1,6k), Pesochen (1,5k) and Zholnin (1,4k), while the southeastern forest-steppe corridor along the Psyol, Vorskla and Seversky Donets hosted Ltava (Poltava) (49,58°N, 34,56°E, 14 ha on *Ivanova Hora*, mentioned 1174AD, 2,2k), Goltav (1,5k), Nizhegolsk (1,4k) and the easternmost frontier fortress of Donets (49,85°N, 36,21°E, Khorosheve hillfort south of modern Kharkiv, where Igor Svyatoslavich escaped from Polovtsian captivity in 1185AD, 2,0k) [19][29].

Lower Dnieper Emporium (Oleshye) and the Polovtsian Steppe Frontier (~210,0k at 1200AD)

South of the Porossia and Posullya ramparts extended the Pontic Steppe (*Pole Polovetskoe*), where 14 fortified river-island and delta trading stations coexisted with the seasonal camps (*vezhi* and winter *zimovniki*) of the Dnieper Polovtsian confederations (the Burchevichi and Lukomorie hordes led by Khans Kobyak, Bonyak and Kotyan) [9]:

- **Oleshye** (46,62°N, 32,62°E, on Veliky Potemkin Island opposite modern Kherson and Oleshky): The maritime gateway of Kievan Rus (20 ha waterlogged port settlement yielding thousands of Byzantine amphorae, lead seals of Kievan princes and Greek officials, glass bracelets and ship fittings), where Kievan military flotillas met Byzantine merchants and Newly consecrated Metropolitans; 3,5k (3 500) permanent inhabitants at 1200AD.
- **Khortytsia** (Protolche) (47,82°N, 35,07°E): The strategic assembly point below the Dnieper Rapids (*Khortytskyi Ostrov* and *Protolche* ford), where the Rus princes mustered before the 1103AD Suten and 1223AD Kalka campaigns (1,8k at 1200AD), alongside Berestovo (Beloberezhye) (1,2k) at the Kinburn spit and Cuman winter towns along the Samara, Orel and Molochnaya rivers (210,0k total sub-regional population at 1200AD).

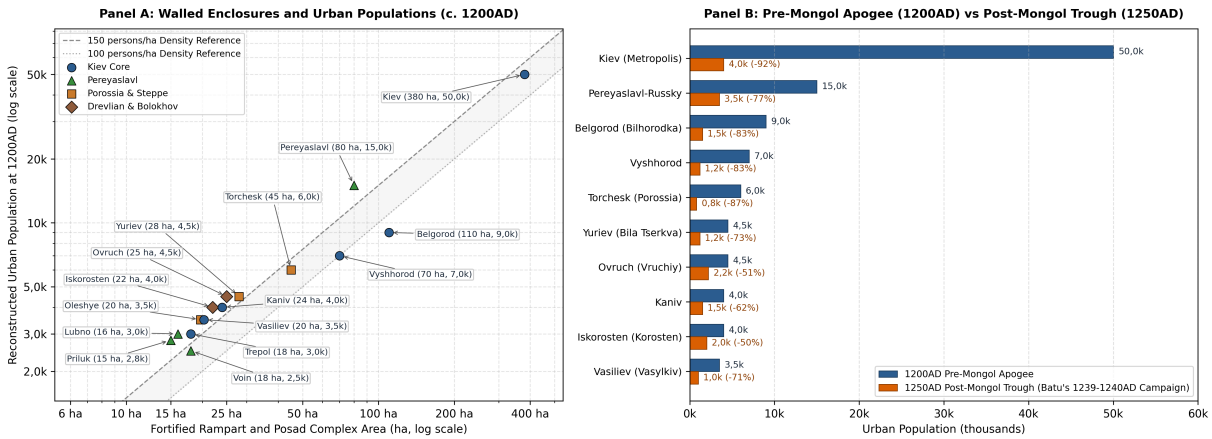


Figure 4. Urban Archaeology and the Demographic Shock of the 1239-1240AD Mongol Sieges

Mongol Conquests and Post-Siege Era (1239AD - 1300AD)

Batu Khan's Campaigns of 1239-1240AD and the Destruction of Southern Rus

Following the initial reconnaissance raid of Jebe and Subutai and the disastrous defeat of the Kievan, Galician, Chernigov and Smolensk princes at the Battle of the Kalka River (31 May 1223AD) - where Grand Prince Mstislav Romanovich of Kiev and 10 000 Kievan warriors perished - the Middle Dnieper principalities entered the 1240AD benchmark with a slightly reduced population of 1,42M (1 420 000). The full-scale Mongol invasion of Southern Rus unfolded in two devastating winter-spring campaigns [6][10][29]:

1. **The Storming of Pereyaslavl and the Left Bank** (March 1239AD): Mongol armies under Möngke and Batu breached the Posullya ramparts, burning Rimiv, Voin, Lubno, Lukoml and Priluk and stormed Pereyaslavl-Russky on 3 March 1239AD. As recorded in the *Laurentian* and *Hypatian Chronicles*, St Michael's Cathedral was demolished, Bishop Simeon was killed and the city's population was slaughtered or led into captivity. Archaeological excavation across the Pereyaslavl *Detinets* and *Posullya* hillforts reveals a continuous 1239AD conflagration horizon containing thousands of Mongol-type rhomboid armour-piercing arrowheads, unburied skeletons in burned cellars and buried treasure hoards (*klady*) that were never recovered [28].
2. **The Siege of Kiev and the Right Bank** (Autumn-December 1240AD): After Möngke's reconnaissance of Kiev from *Pesochen (Gorodok)* on the Left Bank in late 1239AD, Batu Khan's main army crossed the Dnieper in autumn 1240AD, overrunning the Porossia federate towns (Torchesk, Yuriev, Korsun, Vitichev, Vasiliev, Belgorod and Vyshhorod). Defended by Daniel of Galicia's *voivode* Dmitr, Kiev was surrounded by Mongol siege engines (*poroki*), which breached the Lyadskie Gates of Yaroslav's City before a final stand around the Church of the Tithes, which collapsed under the weight of refugees on 6 December 1240AD.

Mass graves excavated by M. K. Karger, P. P. Tolochko and H. Yu. Ivakin in the Kiev Upper City (over 2 000 skeletons in sanitary pits near the Church of the Tithes and St Sophia, alongside intact burned workshops with slain artisans inside) confirm the scale of the urban catastrophe [6][10]. When the Papal envoy Giovanni da Pian del Carpine (John of Plano Carpini) passed through the Kievan land and visited Kiev in February 1246AD, he left an eyewitness account of the post-conquest landscape:

> *"They attacked Rus, where they made great havoc, destroying cities and fortresses and slaughtering men; and they laid siege to Kiev, the capital of Rus; after they had besieged the city for a long time, they took it and put the inhabitants to death. When we were journeying through that land we came across countless skulls and bones of dead men lying about on the ground. Kiev had been a very large and densely populated town, but now it has been reduced almost to nothing, for there are at the present time scarce two hundred houses there and the inhabitants are kept in complete slavery."* [11]

Allowing for Carpini's ~200 taxable household compounds (*dvory*) on *Podol* plus surviving clerical and suburban households around the Caves Monastery and *Kopyrev Konets*, H. Yu. Ivakin reconstructs Kiev's population at 3,5k - 4,5k (4,0k benchmark) in 1250AD - a -92,0% collapse from its 1200AD peak of 50,0k [10].

The Bolokhov Exception (1240-1257AD) and Late 13th-Century Recovery (1250AD - 1300AD)

Archaeological and chronicle evidence shows that demographic loss during the 1239-1240AD invasion was spatially uneven across the Middle Dnieper sub-regions:

- **Severe Depopulation in the Exposed Forest-Steppe** (-64% in Pereyaslavl and Posullya; -55% in Kiev Core and Porossia): Along the Sula and Ros frontier lines, nearly 80% of fortified *gorodishcha* (Voin, Rimiv, Sneporod, Torchesk, Roden, Vitichev) were permanently abandoned after 1239-1240AD [7][28]. At the 1250AD post-conquest trough, the total regional population stood at 740,0k (740 000): 310,0k in the Right-Bank Kiev Core and Porossia, 150,0k in the Drevlian and Bolokhov lands, 150,0k in the Principality of Pereyaslavl and 130,0k in the Lower Dnieper and steppe zone.
- **Resilience in Forested Drevlian Polesia and the Bolokhov Land**: The marshy woodlands of the Teterev-Uzh-Pripyat basin (Ovruch, Iskorosten, Chornobyl, Zhytomyr) sheltered rural refugees fleeing the Dnieper corridor. Simultaneously, the princes and towns of the Bolokhov Land (Bolokhov, Gubin, Kudin, Kobud, Bozhsky) escaped destruction in 1240AD by submitting directly to the Mongols as *Tataropoli* ("Tatar-tribute ploughmen"), sowing wheat and millet for Batu's troops until King Daniel Romanovich of Galicia-Volhynia launched punitive expeditions in 1241AD and 1255-1257AD that razed the Bolokhov ramparts and Vozvyagl [9][27].
- **Golden Horde *Baskak* Administration and 1300AD Benchmark** (820,0k): Following the Golden Horde imperial censuses (*chislo*) of 1257-1259AD and 1273AD under Mengu-Timur, the devolved Principality of Kiev (3368, 3465) and Principality of Pereyaslavl (3350, 3466) were administered by local Rurikid and Putivl-Olgovichi princes alongside Golden Horde *baskaki* (governors and tax-collectors). Although Metropolitan Maximus transferred the practical residence of the Metropolitanate of All Rus from Kiev to Vladimir-on-the-Klyazma in 1299AD, Ivakin's excavations demonstrate steady late 13th-century craft and commercial revival in Kievan *Podol*, *Pechersk* and the castle on *Zamkova Hora* (8,5k inhabitants at 1300AD), Pereyaslavl (4,5k), Ovruch (3,0k), Iskorosten (2,6k), Kaniv (2,2k) and Yuriev (2,0k), bringing the regional population to 820,0k (820 000) at 1300AD prior to the Lithuanian conquest under Gediminas in the 1320s AD [10].

Benchmark	Kiev Core & Porossia (2844, 3368)	Drevlian & Bolokhov Land	Pereyaslavl & Posullya (2848, 3350)	Lower Dnieper & Steppe	Regional Total
500AD	205,0k	80,0k	95,0k	50,0k	430,0k
800AD	310,0k	115,0k	155,0k	100,0k	680,0k
900AD	380,0k	140,0k	195,0k	125,0k	840,0k
1000AD	470,0k	175,0k	255,0k	150,0k	1,05M (1 050,0k)
1100AD	585,0k	215,0k	335,0k	185,0k	1,32M (1 320,0k)
1200AD	690,0k	260,0k	420,0k	210,0k	1,58M (1 580,0k)
1240AD	620,0k	240,0k	370,0k	190,0k	1,42M (1 420,0k)
1250AD	310,0k	150,0k	150,0k	130,0k	740,0k
1300AD	345,0k	165,0k	175,0k	135,0k	820,0k

Table 2. Regional Demographic and Archaeological Data Summary (Table 2).

Cross-Regional Demographic Synthesis (1000AD - 1300AD)

To ensure full demographic consistency across the SVEA historical dataset, the table below cross-references the Middle Dnieper Core (Kiev and Pereyaslavl) with the northern and eastern Rus principalities documented in the companion European Russia regional review [21][23][30].

Polity / Regional Unit	1000AD Pop.	1100AD Pop.	1200AD Apogee	1250AD Trough	1300AD Pop.
Principality of Kiev (2844, 3368, incl. Drevlian & Bolokhov)	645,0k	800,0k	950,0k	460,0k	510,0k
Principality of Pereyaslavl (2848, 3350, incl. Posullya)	255,0k	335,0k	420,0k	150,0k	175,0k
Lower Dnieper (Oleshye) & Pontic Steppe Frontier	150,0k	185,0k	210,0k	130,0k	135,0k
Region 1 Subtotal (kiev_and_pereyaslavl)	1,05M	1,32M	1,58M	740,0k	820,0k
cf. Vladimir-Suzdal / Rostov (european_russia)	340,0k	510,0k	780,0k	520,0k	640,0k
cf. Novgorod & Pskov Republics (european_russia)	290,0k	380,0k	480,0k	460,0k	510,0k

(Continued from previous page)

Polity / Regional Unit	1000AD Pop.	1100AD Pop.	1200AD Apogee	1250AD Trough	1300AD Pop.
cf. Principality of Smolensk (european_russia)	210,0k	275,0k	340,0k	290,0k	310,0k
cf. Kursk-Rylsk & Bryansk-Upper Oka (european_russia)	280,0k	390,0k	500,0k	270,0k	310,0k
cf. Total european_russia Benchmark	2,15M	2,90M	3,85M	2,80M	3,10M

Table 3. Regional Demographic and Archaeological Data Summary (Table 3).

This synthesis highlights the fundamental structural shift of the 12th and 13th centuries: while the Middle Dnieper Core (1,58M at 1200AD) remained the single most densely populated and urbanised territory of the Rus world up to the eve of the Mongol invasion - containing two of the five largest cities (Kiev 50,0k and Pereyaslavl 15,0k) and 240 fortified *gorodishcha* - its open forest-steppe geography left it uniquely exposed to the 1239-1240AD Mongol siege campaigns and subsequent nomadic raids. Whereas the total population of European Russia contracted by -27,3% (3,85M to 2,80M) between 1200AD and 1250AD (buffered by the uninvaded Novgorod-Pskov and Smolensk lands) and recovered to 80,5% of its pre-Mongol peak by 1300AD (3,10M), the Middle Dnieper Core suffered a -53,2% contraction (1,58M to 740,0k in 1250AD) and stood at 51,9% of its pre-Mongol apogee (820,0k) at the 1300AD benchmark.

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